

Glen Ochieng Submission

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Introduction

The packages used were: tidyverse, sf, ggspatial

```
packages= c("tidyverse","sf", "ggspatial")
lapply(packages, library, character.only = T)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE

## [[1]]
## [1] "lubridate" "forcats"    "stringr"    "dplyr"      "purrr"      "readr"
## [7] "tidyr"     "tibble"     "ggplot2"    "tidyverse"  "stats"      "graphics"
## [13] "grDevices" "utils"      "datasets"   "methods"    "base"
##
## [[2]]
## [1] "sf"          "lubridate" "forcats"    "stringr"    "dplyr"      "purrr"
## [7] "readr"       "tidyr"     "tibble"     "ggplot2"    "tidyverse"  "stats"
## [13] "graphics"    "grDevices" "utils"      "datasets"   "methods"    "base"
##
```

```
## [[3]]
## [1] "ggspatial" "sf"          "lubridate" "forcats"    "stringr"    "dplyr"
## [7] "purrr"     "readr"      "tidyr"     "tibble"     "ggplot2"    "tidyverse"
## [13] "stats"     "graphics"   "grDevices" "utils"      "datasets"   "methods"
## [19] "base"
```

Data Importation

```
sle_sf = st_read("who_shapefile_sle_adm2_latest.gpkg")
```

```
## Reading layer 'who_shapefile_sle_adm2_latest' from data source
## 'C:\Users\ADMIN\Desktop\AHADI\Ahadi Assement\who_shapefile_sle_adm2_latest.gpkg'
## using driver 'GPKG'
## Simple feature collection with 14 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -13.30213 ymin: 6.923437 xmax: -10.27168 ymax: 9.999973
## Geodetic CRS: WGS 84
```

```
sle_pop = read.csv("sle_pop_2020_2025.csv")
```

```
glimpse(sle_sf)
```

```
## Rows: 14
## Columns: 7
## $ adm0_code <chr> "SLE", "SLE", "SLE", "SLE", "SLE", "SLE", "SLE", "SLE", "SL~
## $ adm0 <chr> "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEO~
## $ adm1 <chr> "SOUTHERN", "NORTHERN", "SOUTHERN", "EASTERN", "NORTHERN", ~
## $ adm2 <chr> "BO", "BOMBALI", "BONTHE", "KAILAHUN", "KAMBIA", "KENEMA", ~
## $ start_date <date> 2000-01-01, 2000-01-01, 2000-01-01, 2000-01-01, 2000-01-01~
## $ end_date <date> 9999-12-31, 9999-12-31, 9999-12-31, 9999-12-31, 9999-12-31~
## $ geom <MULTIPOLYGON [°]> MULTIPOLYGON (((-11.86631 7..., MULTIPOLYGON (~
```

```
glimpse(sle_pop)
```

```
## Rows: 58
## Columns: 5
## $ adm0 <chr> "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "~
## $ adm1 <chr> "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN",~
## $ adm2 <chr> "KAILAHUN", "KAILAHUN", "KAILAHUN", "KAILAHUN", "KENEMA", "KENEMA~
## $ year <int> 2020, 2021, 2023, 2025, 2020, 2021, 2023, 2025, 2020, 2021, 2023,~
## $ pop <chr> "567,273", "578618", "601994", "626315", "816136", "832459", "866~
```

Data Cleaning

I followed the following steps to clean my data.

1. Replaced the two nulls in adm2 column.

2. For the pop column, I removed the commas, replaced the O's with 0 and deleted trailing 0's to correct the large outlier populations. 3. I dropped the 2 duplicate values: Kambia 2021 and Western Area Rural 2023.
3. Changed the columns to the correct data types.

The codes for the cleaning are as below:

```
unique(sle_pop$adm0) # adm0 seems okay
```

```
## [1] "SIERRA LEONE"
```

```
unique(sle_pop$adm1) #adm1 seems good - just the 4
```

```
## [1] "EASTERN"      "NORTHERN"      "SOUTHERN"      "WESTERN AREA"
```

```
unique(sle_pop$year) # years seem good - just the 4
```

```
## [1] 2020 2021 2023 2025
```

```
unique(sle_pop$adm2) # adm2 seems to have a NA's
```

```
## [1] "KAILAHUN"      "KENEMA"      "KONO"
## [4] "BOMBALI"      "KAMBIA"      NA
## [7] "KOINADUGU"    "PORT LOKO"   "TONKOLILI"
## [10] "BO"          "BONTHE"      "MOYAMBA"
## [13] "PUJEHUN"      "WESTERN AREA RURAL" "WESTERN AREA URBAN"
```

```
unique(sle_pop$pop) # plenty of issues spotted
```

```
## [1] "567,273"      "578618"      "601994"      "626315"      "816136"      "832459"
## [7] "866090"      "901080"      "340632"      "347445"      "361482"      "376086"
## [13] "579583"      "591175"      "615058"      "639906"      "413430"      "421699"
## [19] "438736"      "456461"      "415887000"   "424205"      "441343"      "459173"
## [25] "681528"      "695159"      "723243"      "752462000"   "548648"      "559621"
## [31] "582230"      "605752"      "834700"      "851394"      "885790"      "921576"
## [37] "194415"      "198303"      "206314"      "214,649"     "317037"      "323378"
## [43] "336442"      "350034"      "478012"      "487572"      "507270"      "527764"
## [49] "393388"      "401256"      "417467"      "434,333"     "1189488"     "1213278"
## [55] "1262294"     "1313291"
```

```
# Cleaning the adm2 column
# After checking manually the dataset for context, I can fill the NA with the
# the appropriate values
```

```
sle_pop$adm2[sle_pop$adm1 == "NORTHERN" &
              sle_pop$year == 2023 &
              is.na(sle_pop$adm2)] = "KAMBIA"
```

```
sle_pop$adm2[sle_pop$adm1 == "NORTHERN" &
```

```

sle_pop$year == 2025 &
is.na(sle_pop$adm2)] = "KOINADUGU"

## Cleaning the pop col

# removing the commas
sle_pop$pop = gsub(",", "", sle_pop$pop)

# replacing the 0 with 0
sle_pop$pop = gsub("0", 0, sle_pop$pop)

# correcting the extra large numbers
sle_pop$pop[sle_pop$adm2 == "KOINADUGU" & sle_pop$year == 2020] <- "415887"
sle_pop$pop[sle_pop$adm2 == "PORT LOKO" & sle_pop$year == 2025] <- "752462"

# checking for duplicates basing of the knowledge the adm1,adm2 and year have
# to be unique

sle_pop |>
  group_by(adm1, adm2, year) |>
  summarise(n = n()) |>
  arrange(desc(n)) # anything above n =1 is a duplicate

```

```

## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by adm1, adm2, and year.
## i Output is grouped by adm1 and adm2.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(adm1, adm2, year))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.

```

```

## # A tibble: 56 x 4
## # Groups:   adm1, adm2 [14]
##   adm1      adm2      year      n
##   <chr>    <chr>    <int> <int>
## 1 NORTHERN  KAMBIA      2021      2
## 2 WESTERN AREA WESTERN AREA RURAL 2023      2
## 3 EASTERN   KAILAHUN    2020      1
## 4 EASTERN   KAILAHUN    2021      1
## 5 EASTERN   KAILAHUN    2023      1
## 6 EASTERN   KAILAHUN    2025      1
## 7 EASTERN   KENEMA      2020      1
## 8 EASTERN   KENEMA      2021      1
## 9 EASTERN   KENEMA      2023      1
## 10 EASTERN  KENEMA      2025      1
## # i 46 more rows

```

```

# so far we have 2 duplicates in our dataset : Kambia 2021 and
# Western Area Rural 2023.

```

```

# Dropping the dups

```

```

sle_pop = sle_pop |>

```

```

distinct()

glimpse(sle_pop) # new dataset less 2 rows. Great

## Rows: 56
## Columns: 5
## $ adm0 <chr> "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "~
## $ adm1 <chr> "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN",~
## $ adm2 <chr> "KAILAHUN", "KAILAHUN", "KAILAHUN", "KAILAHUN", "KENEMA", "KENEMA~
## $ year <int> 2020, 2021, 2023, 2025, 2020, 2021, 2023, 2025, 2020, 2021, 2023,~
## $ pop <chr> "567273", "578618", "601994", "626315", "816136", "832459", "8660~

# Lastly , changing year and pop to the correct ones
sle_pop$year = as.integer(sle_pop$year)
sle_pop$pop = as.integer(sle_pop$pop)

# Pop Dataset is clean
glimpse(sle_pop)

## Rows: 56
## Columns: 5
## $ adm0 <chr> "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "~
## $ adm1 <chr> "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN",~
## $ adm2 <chr> "KAILAHUN", "KAILAHUN", "KAILAHUN", "KAILAHUN", "KENEMA", "KENEMA~
## $ year <int> 2020, 2021, 2023, 2025, 2020, 2021, 2023, 2025, 2020, 2021, 2023,~
## $ pop <int> 567273, 578618, 601994, 626315, 816136, 832459, 866090, 901080, 3~

# Checking the shape file
# Checking values are in and no NA's
unique(sle_sf$adm0_code) # seems okay

## [1] "SLE"

unique(sle_sf$adm0) # seems okay

## [1] "SIERRA LEONE"

unique(sle_sf$adm1) # seems okay

## [1] "SOUTHERN"      "NORTHERN"      "EASTERN"      "WESTERN AREA"

unique(sle_sf$adm2) # seems okay

## [1] "BO"           "BOMBALI"       "BONTHE"
## [4] "KAILAHUN"     "KAMBIA"        "KENEMA"
## [7] "KOINADUGU"    "KONO"          "MOYAMBA"
## [10] "PORT LOKO"    "PUJEHUN"       "TONKOLILI"
## [13] "WESTERN AREA RURAL" "WESTERN AREA URBAN"

```

```
unique(sle_sf$start_date) # seems okay
```

```
## [1] "2000-01-01"
```

```
unique(sle_sf$end_date) #seems okay
```

```
## [1] "9999-12-31"
```

```
# checking duplicates  
any(duplicated(sle_sf))
```

```
## [1] FALSE
```

```
# The shape file is clean as well and the data is in the correct format.
```

Merging the datasets

```
glimpse(sle_pop)
```

```
## Rows: 56  
## Columns: 5  
## $ adm0 <chr> "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "SIERRA LEONE", "~  
## $ adm1 <chr> "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN", "EASTERN",~  
## $ adm2 <chr> "KAILAHUN", "KAILAHUN", "KAILAHUN", "KAILAHUN", "KENEMA", "KENEMA~  
## $ year <int> 2020, 2021, 2023, 2025, 2020, 2021, 2023, 2025, 2020, 2021, 2023,~  
## $ pop <int> 567273, 578618, 601994, 626315, 816136, 832459, 866090, 901080, 3~
```

```
sle_complete = sle_sf |>  
  left_join(sle_pop, join_by(adm0,adm1,adm2))
```

Figures and Analysis

Population Growth

```
options(scipen = 10000)
```

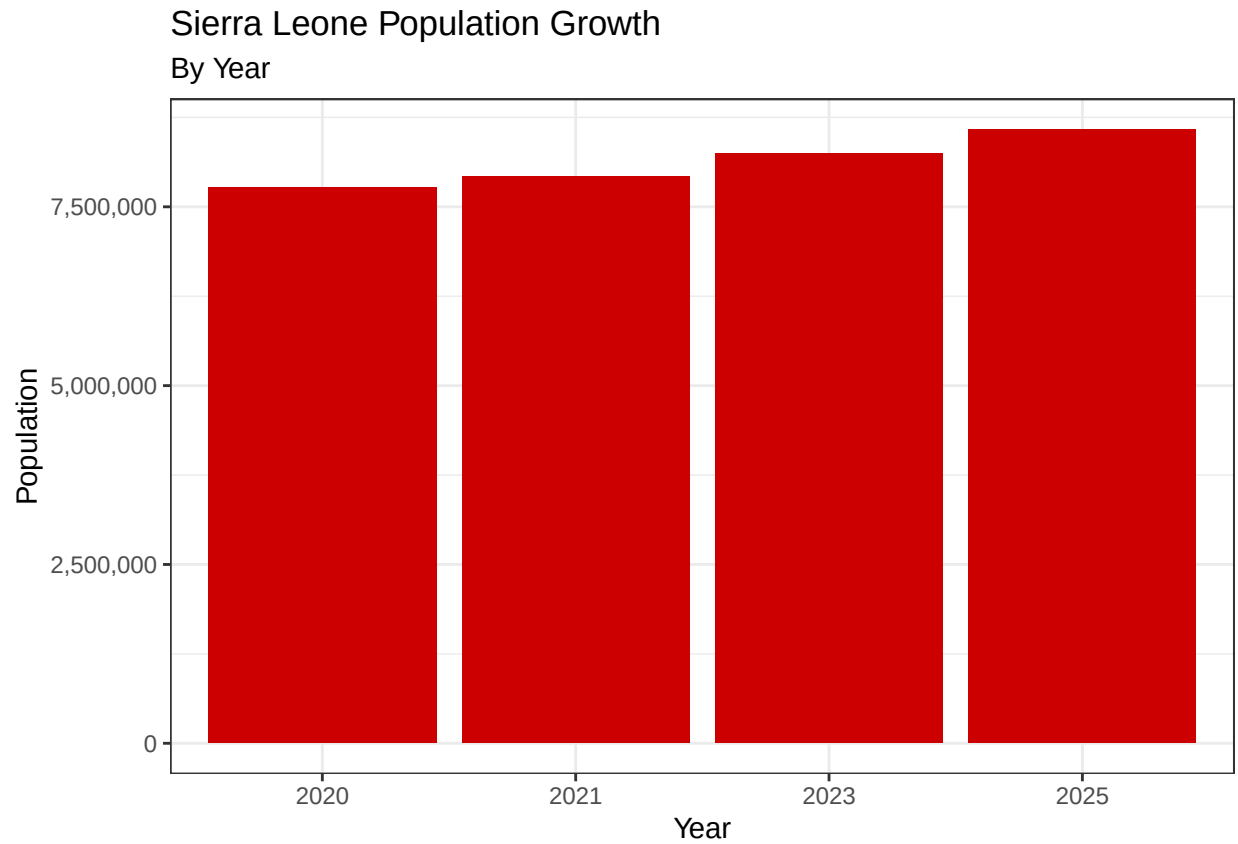
```
## Warning in options(scipen = 10000): invalid 'scipen' 10000, used 9999
```

```
sle_complete |>  
  group_by(year) |>  
  summarise(total_pop = sum(pop)) |>  
  ggplot(aes(x = factor(year), y = total_pop))+  
  geom_col(fill = "red3") +  
  scale_y_continuous(labels = scales::comma) +  
  labs(
```

```

title = "Sierra Leone Population Growth",
subtitle = "By Year",
x = "Year",
y = "Population") +
theme_bw()

```



Sierra Leone's population grew from 7.77 million in 2020 to 8.58 million in 2025; a total increase of 10.4%.

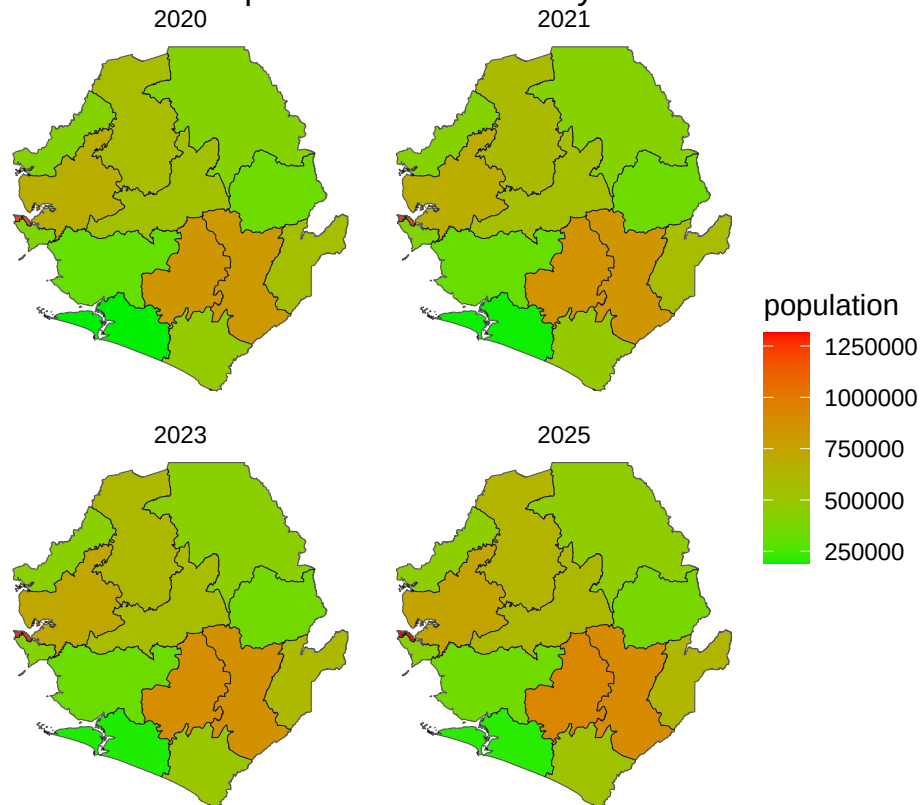
Population Distribution

```

ggplot(sle_complete) +
  geom_sf(aes(fill = pop)) +
  facet_wrap(~year) +
  theme_void() +
  scale_fill_continuous(low = "green2", high = "red") +
  labs(title = "Sierra Leone Population Distribution by Year",
       fill = "population")

```

Sierra Leone Population Distribution by Year



Throughout the years, the NORTHERN REGION has the largest population, approximately 3.4 million more to the second most densely populated region which is the SOUTHERN region.

Western Area Urban, Kenema, and Bo are the three most populous districts, accounting for about 15% the total population